

WORK

Work is defined as the product of the applied nett force and its displacement in the direction of the nett force.

i.e.

$$W = Fs$$

where W = the work done (J-joules)
 F = the nett force applied to the object (N)
 s = the displacement of the object (m)

Work is a **scalar** quantity - no direction is required.

When work is done on an object, it changes the energy of the object.

One **JOULE** of work is done when the point of application of a net force of one newton moves an object through a distance of one metre in the direction of the force.

If the force is applied at an angle to the direction of movement, it is the **component of the force** responsible for the work done.

(This is Year 12 course work and will be covered then.)

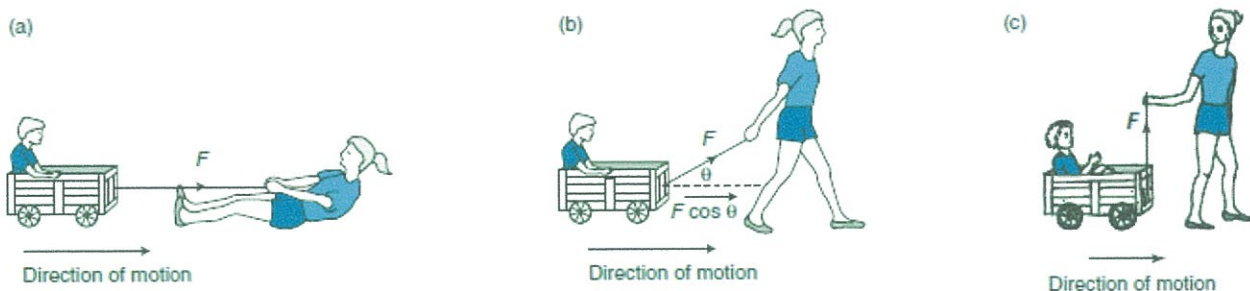


Figure 4.9

(a) If a force is applied in the direction of motion of the cart, then the force is at its most effective in moving the cart. (b) When the force is applied at an angle θ to the direction of motion of the cart, the force is less effective. The component of the force in the direction of the displacement, $F \cos \theta$, is used to calculate the work. (c) When the angle at which the force is acting is increased to a right angle ($\theta = 90^\circ$), then the component of the force in the direction of the intended displacement is zero and it does no work on the cart—provided of course that it doesn't lift the cart, in which case work would also be done against gravity.

If a force is applied to an object and it does not move, then no work has been done.

- e.g. 1. A person may push continuously against a parked car until exhausted, but if the car does not move, **no mechanical work has been done**.

However, the person has expended a significant amount of work internally (physiologically)!

2. Carrying a bag at a constant speed across the ground at a constant height has no work done on it.

Reason: the energy of the bag has not changed.
 (See the section on Energy.)

Examples

1. Calculate the work done against gravity by an athlete of mass 60 kg competing in the Rialto Tower Run-up illustrated in Figure 4.10. Use $g = 9.8 \text{ m s}^{-2}$.

Solution

Only the weight force need be considered in this example as the work in the vertical direction is all that is required.

$$m = 60 \text{ kg}, g = 9.8 \text{ m s}^{-2}, s = \Delta h = 242 \text{ m}$$

$$\text{Force required} = \text{weight} = mg = 60 \times 9.8 = 588 \text{ N}$$

$$W = Fs$$

$$= 588 \times 242$$

$$= 142\,296 \text{ N m} = 1.4 \times 10^5 \text{ J}$$

Consider movement vertically.

$$W = F_{up} s$$

$$= F_w s$$

$$= mgs$$

$$= (60.0)(9.80)(242)$$

$$= 1.42 \times 10^5 \text{ J.}$$



Figure 4.10

The annual Rialto Tower Run-up is a race to climb 1254 steps up the Rialto Tower building in Melbourne. Contestants climb a vertical distance of approximately 240 m and the winner takes around 7 minutes to complete the race.

2. The girl in Figure 4.9 pulls the cart by applying a force of 50 N at an angle of 30° to the horizontal. Assuming a force due to friction of 10 N is also acting on the wheels of the cart, calculate the work done if the cart is moved 10 m along the ground in a straight line.

Solution

$$F_1 = 50 \text{ N}, \theta = 30^\circ, F_f = 10 \text{ N}, s = 10 \text{ m}$$

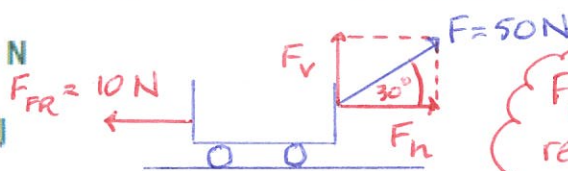
$$\Sigma F = F_1 \cos 30^\circ - F_f$$

$$= 50 \times 0.866 - 10 = 33.3 \text{ N}$$

$$\text{Now } W = \Sigma F s,$$

$$\text{and work} = 33.3 \times 10 = 333 \text{ J}$$

Done in Year 12 course.



$$F_h = 50 \cos 30^\circ = 43.3 \text{ N.}$$

F_h is the force responsible for horizontal movement.

Horizontally: $\longrightarrow +ve$

$$\Sigma F_h = 43.3 - 10.0 = 33.3 \text{ N.}$$

$$W = \Sigma F_h s$$

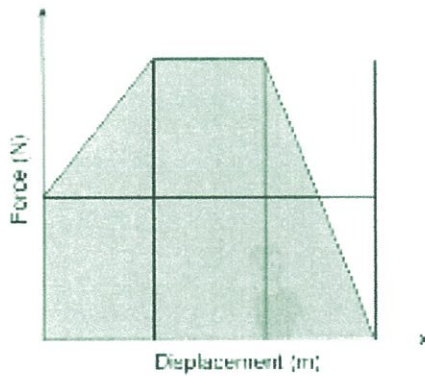
$$= (33.3)(10.0)$$

$$= 333 \text{ J.}$$

Force - Displacement Graphs

The area under such a graph gives a measure of the work done by the force.

This is particularly useful in situations where a force changes with displacement, such as in collisions.



$$\begin{aligned}\text{The area under the graph} &= W_{\text{done}} \\ &= F_{\text{ave}} s\end{aligned}$$

Thus it is possible to determine the average force which acted during the situation being studied.

Remember - Area under graph = (y-axis) $\times$ (x-axis)

ENERGY

Energy is a measure of a body or system's ability to do work.

There are various types of mechanical energy which are applied to movement.

KINETIC ENERGY

This is the energy possessed by a body when it is moving.

It is given by:

$$E_k = \frac{1}{2}mv^2$$

where E_k = the kinetic energy (J)
 m = the mass of the object (kg)
 v = the velocity of the mass (ms^{-1})

If a force acts on a nett body, it does work on it. In doing so there is a change in the kinetic energy of the body.

$$\text{i.e. } W = \Delta E_k = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

where $\frac{1}{2}mv^2$ = the final kinetic energy of the body.
 $\frac{1}{2}mu^2$ = the initial kinetic energy of the body.

NOTE

If the final E_k is smaller than the initial E_k of the object because it is slowing down, reverse the equation to produce a positive value for ΔE_k .

$$\text{i.e. } W = \Delta E_k = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$$

Like all forms of energy, E_k is a scalar and has no direction associated with it.

EXAMPLE

A car of mass $1.30 \times 10^3 \text{ kg}$ is moving along a road at 70.0 kmh^{-1} when it brakes sharply to avoid a person on a skateboard coming out of a driveway. The speed is reduced to 30.0 kmh^{-1} as the car swerves around the person. Calculate the work done by the brakes on the car.

$$u = 70.0 \text{ kmh}^{-1} \\ = 19.44 \text{ ms}^{-1}$$

$$v = 30.0 \text{ kmh}^{-1} \\ = 8.333 \text{ ms}^{-1}$$

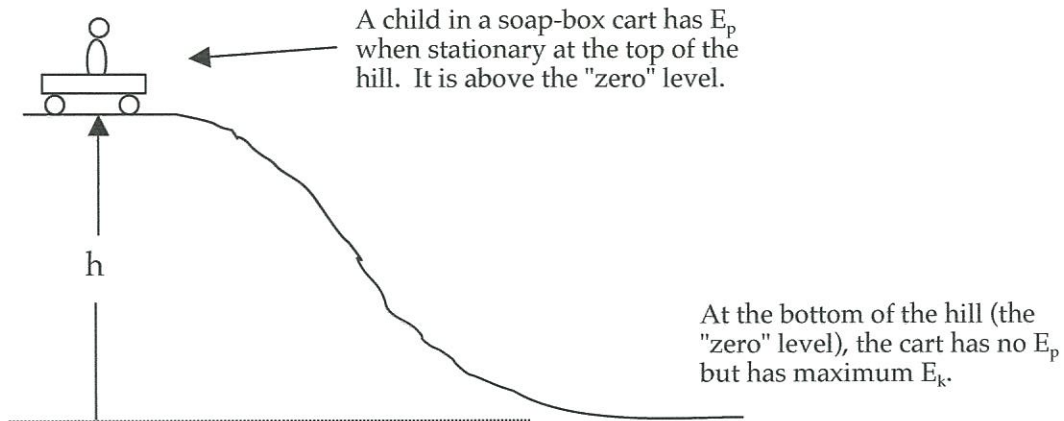
$$W = \Delta E_k = \frac{1}{2}mu^2 - \frac{1}{2}mv^2 \\ = \frac{1}{2}m(u^2 - v^2) \\ = \frac{1}{2}(1.30 \times 10^3) [(19.44)^2 - (8.333)^2] \\ = 2.005 \times 10^5 \text{ J}$$

$$\therefore \underline{W_{\text{done}} = 2.00 \times 10^5 \text{ J}}$$

GRAVITATIONAL POTENTIAL ENERGY

This is the energy possessed by a body due to its height above a given level.

Any object above a given "zero" level has the ability to release this energy as E_k as it falls to this "zero" level.



The amount of potential energy is given by:

$$E_p = mgh$$

where E_p = the potential energy of the body (J)
 m = the mass of the body (kg)
 g = the acceleration due to gravity (ms^{-2})
 h = the height above the "zero" level (m)

If an object is raised up to a height in the Earth's gravitational field, work must be done against the force of gravity.

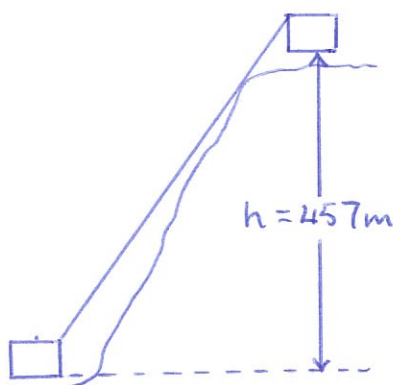
This work is given by:

$$W = \Delta E_p = mg\Delta h$$

where Δh = the height through which the object is raised.

EXAMPLE

A cable car of combined mass $4.30 \times 10^3 \text{ kg}$ is used to transport 22 people from the carpark to the top of Grouse Mountain in Vancouver, a vertical rise of 457 m. Calculate the work done by the motors operating the car.



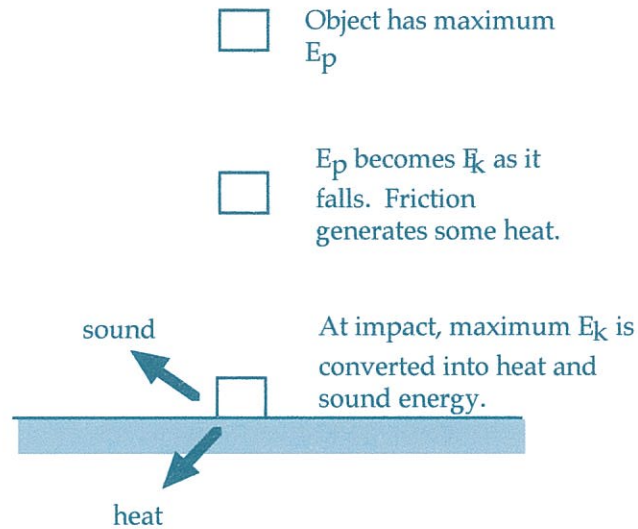
$$\begin{aligned} W_{\text{done}} &= \Delta E_p = mg\Delta h \\ &= (4.30 \times 10^3)(9.80)(457) \\ &= 1.926 \times 10^7 \text{ J} \\ \therefore W_{\text{done}} &= \underline{1.93 \times 10^7 \text{ J}} \end{aligned}$$

CONSERVATION OF MECHANICAL ENERGY

In all situations, energy must be conserved - it can be transformed from one form to another, but the total amount stays the same.

EXAMPLE

Consider an object dropped from a height onto a hard surface.



The above situation describes a particular case where the total mechanical energy is conserved.

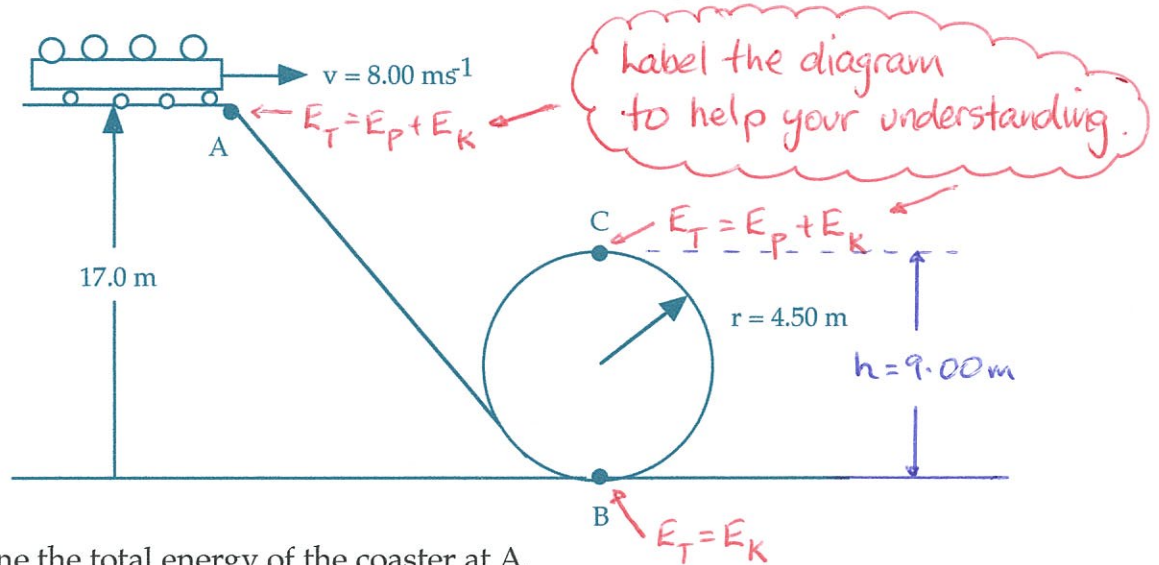
$$\begin{aligned} E_{\text{total}} &= E_p + E_k \\ &= mgh + \frac{1}{2}mv^2 \end{aligned}$$

NOTE

In calculations, it will generally be considered that friction is not acting.

EXAMPLE

A roller coaster of mass $3.70 \times 10^3 \text{ kg}$ is moving at 8.00 ms^{-1} at point A as shown below.



- Determine the total energy of the coaster at A.
- Calculate the speed achieved at point B.
- Does the coaster safely make it through the loop if the minimum speed to do so is 6.64 ms^{-1} ? Justify your answer by calculation.

NOTES

Stationary at a height $\Rightarrow E_T = E_P$

Moving at a height $\Rightarrow E_T = E_P + E_K$

Moving at lowest level $\Rightarrow E_T = E_K$

(a) At A: Moving at a height

$$\begin{aligned}\therefore E_T &= E_P + E_K \\ &= mgh + \frac{1}{2}mv^2 \\ &= (3.70 \times 10^3)(9.80)(17.0) + \frac{1}{2}(3.70 \times 10^3)(8.00)^2 \\ &= 7.348 \times 10^5 \text{ J} \\ \therefore \underline{E_T} &= \underline{7.35 \times 10^5 \text{ J}}\end{aligned}$$

(b) At B: Moving at lowest level.

$$\begin{aligned}\therefore E_T &= E_K = \frac{1}{2}mv^2 \\ \Rightarrow 7.348 \times 10^5 &= \frac{1}{2}(3.70 \times 10^3)v^2 \\ \Rightarrow v &= 19.93 \text{ ms}^{-1} \\ \therefore \underline{\text{Speed}} &= \underline{19.9 \text{ ms}^{-1}}\end{aligned}$$

(c) At C: Moving at a height.

$$\begin{aligned}\therefore E_T &= E_P + E_K \\ &= mgh + \frac{1}{2}mv^2 \\ \Rightarrow 7.348 \times 10^5 &= (3.70 \times 10^3)(9.80)(9.00) + \frac{1}{2}(3.70 \times 10^3)v^2 \\ \Rightarrow v &= 14.86 \text{ ms}^{-1} (> 6.64 \text{ ms}^{-1}) \\ \therefore \underline{\text{The coaster safely negotiates the loop.}}\end{aligned}$$

ELASTIC AND INELASTIC COLLISIONS

In all collisions, momentum is conserved. So too is energy (in all its different forms). However, it is unusual for a particular type of energy to be conserved during a collision.

ELASTIC COLLISION - one where no kinetic energy is lost, and the E_k and momentum of the system are both conserved.

e.g. In the interaction between atoms and sub-atomic particles.

Elastic collisions do not exist in everyday situations - only the sub-atomic.

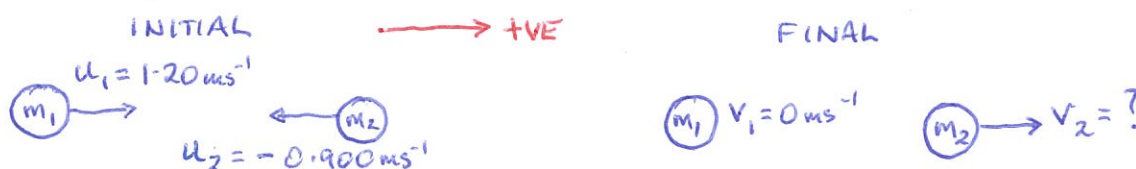
INELASTIC COLLISION - one where momentum is conserved but not E_k (which is transformed into other forms of energy).

e.g. Collisions between billiard balls, cars, football players, etc.

EXAMPLE

A billiard ball of mass 0.150 kg moving at 1.20 ms^{-1} collides head-on with a similar ball moving at 0.900 ms^{-1} in the opposite direction. After the collision the first ball becomes stationary.

- Calculate the final velocity of the second ball.
- Compare the sum total of the initial and final kinetic energies.



- Momentum is conserved in mechanical systems.

$$\sum p_i = \sum p_f$$

$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\Rightarrow (0.150)(1.20) + (0.150)(-0.900) = 0 + (0.150)v_2$$

$$\Rightarrow v_2 = 0.300 \text{ ms}^{-1}$$

$\therefore v_2 = 0.300 \text{ ms}^{-1}$ in the direction of ball 1.

$$\begin{aligned} (b) \quad \sum E_k(\text{initial}) &= \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \\ &= \frac{1}{2} (0.150)(1.20)^2 + \frac{1}{2} (0.150)(0.900)^2 \\ &= 0.1688 \text{ J} \end{aligned}$$

$$\begin{aligned} \sum E_k(\text{final}) &= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \\ &= \frac{1}{2} (0.150)(0) + \frac{1}{2} (0.150)(0.300)^2 \\ &= 6.75 \times 10^{-3} \text{ J} \end{aligned}$$

$$\therefore \underline{\sum E_k(\text{initial}) \neq \sum E_k(\text{final})}$$

POWER

POWER - the rate at which energy is transformed, or the rate at which work is done.

$$P = \frac{W}{t} = \frac{E}{t}$$

where P = the power generated (W - Watts)
 W = the work done (J)
 E = the energy transformed (J)
 t = the time taken for the process (s)

Calculating the power developed is straightforward when there is a change in mechanical energy or work, but some situations involve movement at a constant speed which involves no change in mechanical energy.

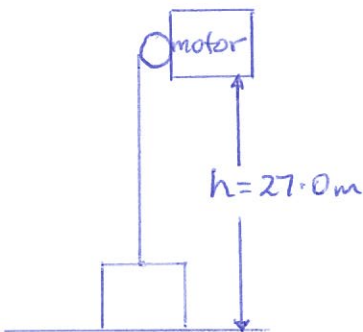
Using $W = Fs$, the equation changes as follows.

$$P = \frac{W}{t} = \frac{Fs}{t} = F v_{av}$$

where F = average force applied (N)
 v_{av} = the average velocity of the object (ms^{-1})

EXAMPLES

1. An electric motor drives a lift of nett mass $1.47 \times 10^4 \text{ kg}$ through a height of 27.0 m in 12.0 s. What is the power of the motor?



$$\begin{aligned} P &= \frac{W}{t} = \frac{\Delta E_p}{t} = \frac{mg\Delta h}{t} \\ &= \frac{(1.47 \times 10^4)(9.80)(27.0)}{12.0} \\ &= 3.241 \times 10^5 \text{ W} \\ \therefore P &= 3.24 \times 10^5 \text{ W} \end{aligned}$$

2. A cyclist in a road race is producing an average of $5.00 \times 10^2 \text{ W}$ of power whilst moving at 48.0 kmh^{-1} along a flat road. Ignoring any air resistance, what average force is being supplied through the pedals?

$$\begin{aligned} V_{\text{const}} &= 48.0 \text{ kmh}^{-1} \\ &= 13.33 \text{ ms}^{-1} \end{aligned}$$

$$P = \frac{W}{t} = \frac{Fs}{t} = Fv_{\text{ave}}$$

$$\Rightarrow F = \frac{P}{v_{\text{ave}}}$$

$$= \frac{5.00 \times 10^2}{13.33}$$

$$= 37.51 \text{ N}$$

$$\therefore \underline{F = 37.5 \text{ N forwards}}$$